

How can slip stick friction be maximized between finger and computer mice

Jimmy Zeng

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1 Introduction

Friction is a topic that has intrigued people for millenia, and pioneered to this very day. Think honestly, what causes friction? Small microscopic asperities? How does perfectly smooth rubber still present friction? Or jagged ice is so slippery? Is water the issue? Then why can you tread on water? From scientists, control engineers, and frankly, scholars that experienced its' effects in other forms, all trying to understand this confounding concept. Namely, slip-stick friction is the limit cycle of an objects' continuous oscillation between slipping and sticking, such as seen in violins, door's creakings, badly engineered anything. It is a topic fixated by engineers in trying to reduce, common solutions include adding grease, oil, etc. However, in the realm of competitive Minecraft, there exists many instances where the desirable trait of clicking fast can be accompanied by exploitation of slip stick atop the mouse. This is referred to in the community as "drag clicking".

In this paper, we will attempt to find ways to test the slip stick properties pertaining to drag clicking by modeling material properties of common methods using advanced friction models such as LuGre and Dahl friction models.

2 Background

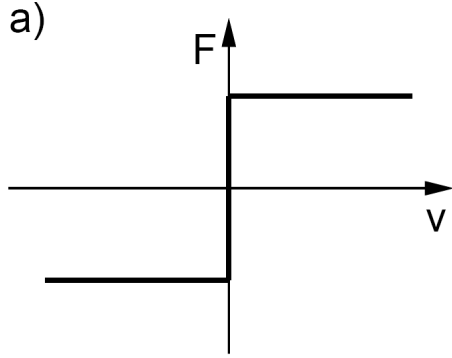


Figure 1: The change of friction with respect to velocity

Our commonly conceived model of friction—the coulomb friction model—has several limitations. It suggests that upon motion the frictional force is linearly proportional to the normal force and of the opposite direction of the velocity [4]. illustrated in figure 1. This is perceived as due to microscopic asperities interlocking with one another. This, for instance explains the independence of apparent surface area and dependence on normal force (i.e. the normal force pushes more asperities in contact, despite small appearant contact area, true microscopic contact area is much larger due to the increased force) However, friction still has much greater depth covering phenomena such as:

1. The Stribeck Effect
2. The Microscale pre-slip deformations
3. Dependence on past events
4. Ultimate Friction aging

2.1 Friction Phenomena

2.1.1 Viscous Friction

Viscous friction introduces the role of velocity in that sliding friction increases with velocity. Though exact instantaneous velocity cannot be provided via discrete data points, a treadmill setup could mitigate this with a constant velocity upon contact

2.1.2 Static Friction

Friction exhibits a noticable decrease when set in motion. Increasing applied force beyond this is usually the exciter for slip-stick friction.

2.1.3 The Stribeck Effect

The Stribeck Effect suggests that static friction does not suddenly jump to viscous friction but happens gradually. The rate of friction mode change depends much on material properties and in particular, we expect sharper jumps to exhibit superior drag-clicking capabilities

2.1.4 Rising Static friction

Refers to the increase in static friction as a result of time. Described as the normal forces gradually increasing true surface area between microscopic asperities over time.

2.2 State Space Equations

To coherently account for behaviour involving state changes, we must shift to dynamic models that uses state space equations. Most notably, the Dahl Model and LuGre Model. In state space equations, there exists "state variables" that capture internal states of the system, when there is more than one, it is packaged into a vector matrix. State space equations define the derivative of state variable as a function of state variables, this gives us the rate of change at an instant, as we integrate it over small discrete units of time it gives us a prediction of the current state (current position, force, etc.). The predicted state variables are fed back into the derivative as linear equation constants to be then integrated again.[5] However, looking at the relationship between variables in a state space equation can describe the stability of a system[5], which is particularly useful in stick-slip motion.

2.3 The Dahl Model

While the standard textbook definition for friction stem from rigid body microscopic asperities, the Dahl Model instead visualizes two bodies making contact from elastic bristles scattered over the contacting surface[2]. Illustrated in figure 2.

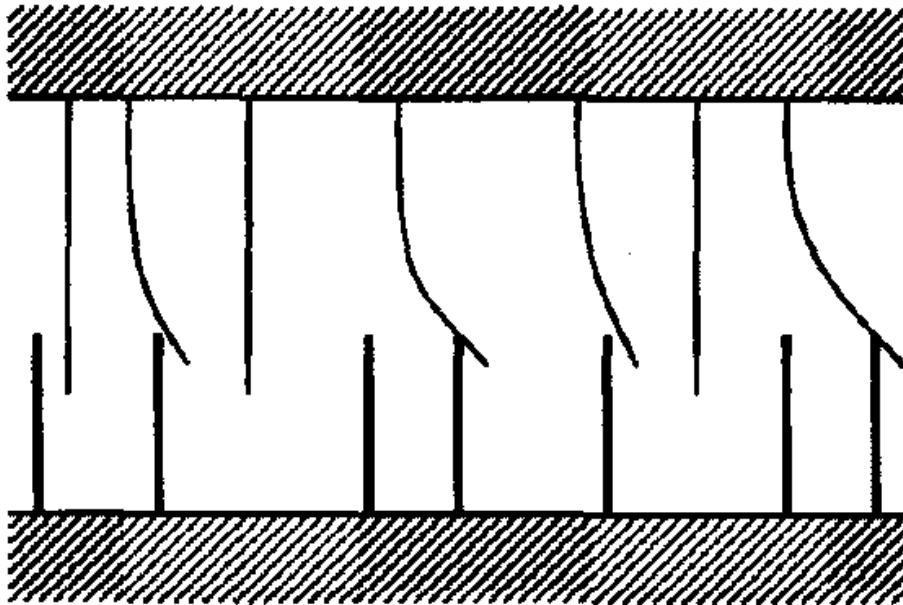


Figure 2: Depiction of Dahl Model's Bristles [2]

The Dahl model is defined as the following:

$$F = F_c \left(1 - e^{-\frac{\sigma_0 |x|}{F_c}} \right) \operatorname{sgn} \left(\frac{dx}{dt} \right) \quad (1)$$

Where F_c is the coulomb friction of the surface, x is the displacement from original position (this is primarily focused on the microscopic "pre-displacements"), and σ_0 is the "stiffness" of microscopic bumps and affects the shift to maximum coulomb friction.

To use this for state space equations we must find the derivative using chain rule:

$$\frac{dF}{dx} = \cancel{\frac{F}{F_c}} \cdot e^{-\frac{\sigma_0|x|}{F_c}} \cdot \cancel{\frac{\sigma_0}{F_c}} \operatorname{sgn}\left(\frac{dx}{dt}\right) \operatorname{sgn}(x) \quad (2)$$

$$= \sigma_0 e^{-\frac{\sigma_0|x|}{F_c}} \operatorname{sgn}\left(\frac{dx}{dt}\right) \operatorname{sgn}(x) \quad (3)$$

and to put the actual current force state back into this derivative function (since state space equations depend on previous states), we rearrange eq (1) to isolate $e^{-\frac{\sigma_0|x|}{F_c}}$ like such:

$$F = F_c \left(1 - e^{-\frac{\sigma_0|x|}{F_c}}\right) \operatorname{sgn}\left(\frac{dx}{dt}\right) \quad (4)$$

$$\frac{F}{F_c} \operatorname{sgn}\left(\frac{dx}{dt}\right) = 1 - e^{-\frac{\sigma_0|x|}{F_c}} \quad (5)$$

$$e^{-\frac{\sigma_0|x|}{F_c}} = 1 - \frac{F}{F_c} \operatorname{sgn}\left(\frac{dx}{dt}\right) \quad (6)$$

and putting it back in the differential equation we get:

$$\frac{dF}{dx} = \sigma_0 \left(1 - \frac{F}{F_c}\right) \operatorname{sgn}\left(\frac{dx}{dt}\right) \operatorname{sgn}(x) \quad (7)$$

$$= \sigma_0 \left(1 - \frac{F}{F_c}\right) \operatorname{sgn}(x) \quad (8)$$

The Dahl model's equation shares similar properties of an exponential function take of the stress-strain curve (depicted in figure 3), most notably, the point at which it shifts from an elastic spring to continual bristle movement

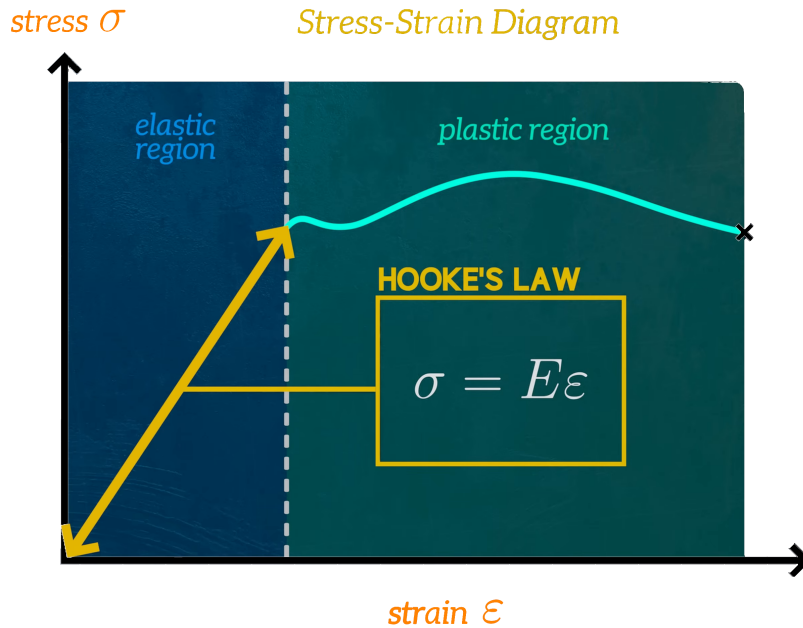


Figure 3: An annotated stress-strain curve[3]

Most importantly the Dahl Model introduces $z = \frac{F}{\sigma_0}$ as a state variable that represents the "average bristle deflection" [46]. This is very similar to Hooke's law. rearranging, we get:

$$F = \sigma_0 z \quad (9)$$

similarly, in Hooke's law presented in the Physics formula booklet:

$$F = kx \quad (10)$$

Differentiating $z = \frac{1}{\sigma_0} \cdot F$ to find the state equation:

$$\frac{dz}{dt} = \frac{1}{\sigma_0} \cdot \frac{dF}{dx} \cdot \frac{dx}{dt} \quad (11)$$

$$= \frac{1}{\sigma_0} \cdot \frac{dF}{dx} \cdot v \quad (12)$$

$$(13)$$

and with $\frac{dF}{dx} = \sigma_0 \left(1 - \frac{F}{F_c} \text{sgn}(v)\right)$

$$= \frac{1}{\sigma_0} \cdot \sigma_0 \left(1 - \frac{F}{F_c} \text{sgn}(v)\right) v \quad (14)$$

$$= \left(1 - \frac{F}{F_c} \text{sgn}(v)\right) v \quad (15)$$

$$(16)$$

now with $F = \sigma_0 z$ and $\text{sgn}(v)v = |v|$ we arrive at the definitive Dahl's model

$$= \left(1 - \frac{\sigma_0 z}{F_c} \text{sgn}(v)\right) v \quad (17)$$

$$\frac{dz}{dt} = v - \frac{\sigma_0 z}{F_c} |v| \quad (18)$$

[4]

2.4 Lugre Model

The Lugre model takes in consideration that coulomb friction is not constant but rather dependent of velocity, thereby replacing it with $g(v) = F_c + (F_s - F_c) e^{-|\frac{v}{v_s}|^\alpha}$ that considers the rate of change from F_s to F_c to be non-instantaneous as shown in figure 4

to create the following differential equation:

3 Experiment

3.1 Drag clicking specific methods

There exists several known methods in the community to increase the accessibility of drag clicking on a mouse. With any youtube search you will find it populated with applying "tape" to your mouse, along with sharpie markers applied to that, then the mention of

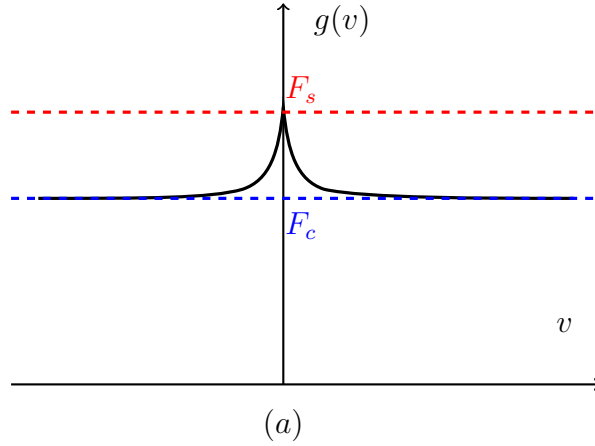


Figure 4: Graph of $g(v)$ with relative F_s and F_c coordinates

keeping your hands dry. In private I have discovered several new methods to increase friction, along with additive methods, I have discovered subtractive methods. Additive methods include rubbing crushed cello rosin dust onto the surface of the mouse with your finger, it was meant to adjunct slip stick properties of stringed instruments, though the effect is universal. Subtractive include rubbing the surface of your mouse with the bottom of your mousepad firmly, or with the leaf of a well hydrated plant. We will measure the rate at which static friction equalizes to coulomb friction with respect to time

3.2 Research Hypothesis and Variables

The difference between Coulomb friction and static friction changes depending on material surface. This investigation aims to determine how material surface affects the difference between coulomb and static coefficient of friction. It is hypothesized that Cello rosin will have the largest change due to its sticky nature, and clear PC plastic is expected with the least coefficient of friction as the surface is factory new surface polished. The independent variables are clear pc, cello rosin, and sharpie while the independent variable is the difference between static and coulomb coefficient of friction.

3.3 Method

The static and kinetic friction will be found by measuring steady friction force at various velocities [4]. The following diagram shows the setup:

A finger sits on a scale while taped to a string horizontally, which passes through a pulley upwards connecting to force gauge with a force exerted by a motor (Vex V5 Smart Motor 11W). The reason it is upwards is because if it were downwards, the weight of the force gauge would inherently exert a force due to gravity despite no applied force from the motor pulling it down and still displaying zero.

The experimental procedure upon procuring this setup is:

1. Clean the surface with alcohol, or reapply rosin, sharpie will not be reapplied upon initial application
2. Clean the finger with alcohol, or reapply rosin, again sharpie is neglected
3. Tare the scale and while at null force, tare the force gauge

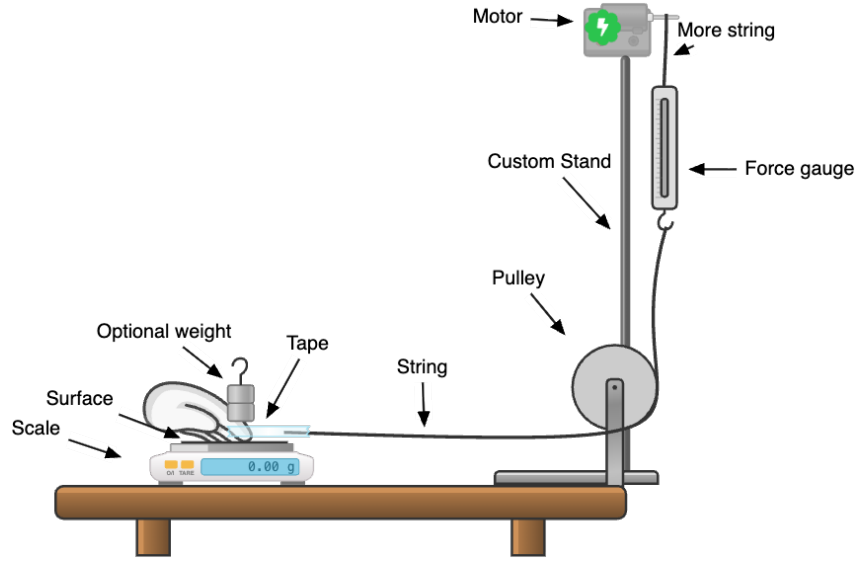


Figure 5: The experimental setup for measuring Coulomb and static friction

4. Insert the string into the pulley
5. Place finger onto the scale
6. While recording slow motion, ensure visibility:
 - (a) Finger state (slip/stick)
 - (b) Force gauge value
 - (c) Scale value
7. Start the motor
8. repeat steps 1-8 3 trials
9. repeat for different materials

4 Results and Discussion

4.1 Observations

The electrical tape was unable to enter kinetic friction, this is likely due to the fact that that the α in the lugre model is too large, meaning the transition to coulomb friction was too rapid (i.e. any increase in velocity resulted in a rapid decrease of frictional force to keep from sliding). It could be possible for the finger to indeed be in motion, except that the velocity is close to zero, explaining the seeming impossibility of reaching visible coulomb friction. Since the tape is viscoelastic, it has a strong elastic spring property, that like described in figure 3, the resultant frictional force is directly proportional to displacement, of which is a result of applied force, which explains the difficulty in achieving kinetic friction, and once the applied force reaches the breaking point, it slips. the addition of the elasticity of the finger introduces additional difficulty in reaching kinetic coulomb friction since elastic on elastic contact gives extra opportunity for the reaction force to reflect the applied force.

For finger systems, it could be suggested that a modified lugre model should be used in which the σ_0 in $z = \frac{F}{\sigma_0}$ reflects the spring constant of a finger rather than microscopic bristles.

The trials of rosin exhibited slip stick of short slip distance before immediately sticking and slipping again. This could be a result of the material being non-viscoelastic, as such, the friction did not have the double spring effect, otherwise it would have doubled the spring of finger and material's displacement before finally being able to slip. Instead, it immediately jumped to coulomb in the case of sufficient applied force. This would be the result of a high α in lugre model's variation of coulomb friction as a result of velocity described in $g(v)$. And this could be a result of particulate friction that has not been looked into in this research, but the stickiness of rosin dust on the finger could be responsible for replacing the skin sebum from contacting the surface of the rosin on the pc board, and instead change into rosin on rosin contact.

4.2 Limitations of the data

The force sensor used was for measuring masses and would stop updating when no change is detected. This significantly impacted the accuracy of the data. Further improvement in methodology could involve high frequency force meter and scale that directly output into computer for high frequency data collection, in particular a computer with high monitor refreshrate and subsequently a better slow motion camera can capture much more details entailing the phases of stick slip behavior. Another limitation is the subject's muscle contraction when resetting, it is possible that force is applied prior to place the finger at the end for the string to not fall out of the pulley. It is also possible for the subject to unintentionally produce opposite force in the same direction of the frictional force to maximize slip stick behavior. Perhaps a future experiment with a moving surface rather than a moving finger would be used instead similar to [1]

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